Exact single-use Bayesian Nagaoka–Hayashi attainment for noisy qubit rotations

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We determine the exact Bayesian precision for simultaneous estimation of three qubit-rotation coordinates using one fixed Bell probe and a noiseless ancilla. The rotation is followed by known depolarization of strength $0 \le \lambda \le 1$, the prior is uniform Cartesian volume on the radius-$\pi/4$ parameter ball, and the loss is one third of the squared coordinate error. The optimum over all joint measurements and estimators equals the Bayesian Nagaoka–Hayashi relaxation and is $\pi^2/80 - 4(1-\lambda)^2[3/(2\pi) - 12/\pi^3]^2$. Four orthogonal maximally entangled outcomes attain this value: a fixed local rotation followed by a Bell measurement suffices, while the posterior estimates shrink with noise. A covariant sphere measurement is also optimal. We give algebraic positive-semidefinite certificates for both the physical risk and the relaxation, valid throughout the noise interval, including its pure-state and no-information endpoints. The result converts the previous single-use numerical near-attainment into a constructive exact precision limit. It concerns the fixed single-use experiment; attainment for two through four parallel uses is not established.

I. INTRODUCTION

Prior information and measurement incompatibility both matter when only a finite number of quantum observations are available. For quadratic Bayesian loss, prior moments determine the optimal posterior estimator and a symmetric-logarithmic-derivative (SLD) lower bound [1]. The Nagaoka–Hayashi (NH) semidefinite formulation [2] has a Bayesian extension [3] that can be tighter, but the relaxed measurement moments need not by themselves specify a physical measurement. Recent work relates this distinction to Bayesian measurement incompatibility [4].

Bia and Gagatsos studied simultaneous estimation of all three qubit-rotation coordinates with a uniform ball prior and depolarizing noise [5]. For up to four parallel uses, their computed strategies nearly attain the Bayesian NH bound, both with optimized probes and with fixed tensor products of Bell probes. They explicitly characterize this as numerical evidence rather than an attainment theorem. We settle the fixed-Bell, single-use sector exactly for every allowed noise strength. The contribution is a finite attaining measurement and matching algebraic certificates, together with a closed precision formula.

The scalar-plus-triplet prior moments used below have a close counterpart in the phase-imaging example of Ref. [4], whose NH attainment is also assessed numerically. This representation and the general Bayesian optimality framework are established tools. Likewise, NH-saturating noisy-rotation measurements in a local, unbiased, phase-damping model were constructed in Ref. [2]. Our result concerns a finite volume prior, depolarization, and posterior estimation without an unbiasedness condition. We do not extend it to multiple uses, other probes, other losses, or adaptive experiments.

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II. EXPERIMENT AND EXACT PRIOR MOMENTS

The system qubit and an untouched ancillary qubit are initially in $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The system undergoes

$$U_{\boldsymbol{\theta}} = e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\sigma}}, \qquad \mathcal{D}_\lambda(\rho) = (1-\lambda)\rho + \lambda \mathbb{I}_2/2, \tag{1}$$

with known $\lambda \in [0,1]$. The Pauli eigenvalues are ± 1; there is no factor $1/2$ in the exponent. The Bloch-vector rotation angle is consequently $2\|\boldsymbol{\theta}\|$. In system–ancilla order the normalized output is

$$\rho_{\lambda,\boldsymbol{\theta}} = (1-\lambda)|v_{\boldsymbol{\theta}}\rangle\langle v_{\boldsymbol{\theta}}| + \lambda \mathbb{I}_4/4, \tag{2}$$

$$|v_{\boldsymbol{\theta}}\rangle = (U_{\boldsymbol{\theta}} \otimes \mathbb{I}_2)|\Phi^+\rangle. \tag{3}$$

This is a trace-one state from a completely positive trace-preserving channel. In the input–output Choi order of Ref. [5], if W swaps the factors, $\rho_{\lambda,\boldsymbol{\theta}} = W J_{\boldsymbol{\theta}} W^\dagger/2$; the unnormalized one-use Choi operator has trace two.

Put $a = \pi/4$. The prior and Bayes risk are

$$p(\boldsymbol{\theta}) = \frac{3}{4\pi a^3}\, \mathbf{1}_{\{\|\boldsymbol{\theta}\| \le a\}}, \tag{4}$$

$$R(M, \hat{\boldsymbol{\theta}}) = \frac{1}{3}\mathbb{E}\big[\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2\big]. \tag{5}$$

The expectation includes the prior and measurement outcomes. We allow arbitrary joint positive operator-valued measures (POVMs), finite or continuous, and measurable estimators with finite second moment. All coordinates and costs are dimensionless. The prior is uniform Cartesian volume, with radial density $3r^2/a^3$; it is neither Haar measure nor a uniform radial density. Its no-information risk is

$$R_{\mathrm{p}} = \frac{\mathbb{E}\|\boldsymbol{\theta}\|^2}{3} = \frac{a^2}{5} = \frac{\pi^2}{80}. \tag{6}$$

For a transparent real-coordinate representation define

$$|0\rangle = |\Phi^+\rangle, \qquad |\widetilde{i}\rangle = -i(\sigma_i \otimes \mathbb{I}_2)|\Phi^+\rangle \quad (i = 1,2,3). \tag{7}$$

These four physical vectors are orthonormal. For $\boldsymbol{\theta} = r\boldsymbol{n}$,

$$|v_{\boldsymbol{\theta}}\rangle = \cos r|0\rangle + \sin r \sum_i n_i|\widetilde{i}\rangle. \tag{8}$$

The rephasing in Eq. (7) changes the basis, not the sign of the estimated parameter. Write $E_{ab} = |a\rangle\langle b|$ in this basis, with indices $a, b = 0, 1, 2, 3$, and define

$$P_0 = E_{00}, \quad P_1 = \sum_{i=1}^{3} E_{ii}, \quad F_i = E_{0i} + E_{i0}. \tag{9}$$

The angular identities $\mathbb{E}[n_i] = 0$ and $\mathbb{E}[n_i n_j] = \delta_{ij}/3$ give

$$\Gamma_0 = \int p(\boldsymbol{\theta})\rho_{\lambda,\boldsymbol{\theta}}\, d^3\boldsymbol{\theta} = AP_0 + BP_1, \tag{10}$$

$$\begin{aligned} \Gamma_i &= \int p(\boldsymbol{\theta})\theta_i\rho_{\lambda,\boldsymbol{\theta}}\, d^3\boldsymbol{\theta} = QF_i, \\ A &= (1-\lambda)C + \lambda/4, \\ B &= (1-\lambda)(1-C)/3 + \lambda/4, \\ Q &= (1-\lambda)q, \qquad A + 3B = 1. \end{aligned} \tag{11}$$

The radial integrals evaluate exactly to

$$C = \frac{3}{a^3}\int_0^a r^2\cos^2 r\, dr = \frac{1}{2} + \frac{3}{\pi} - \frac{24}{\pi^3}, \tag{12}$$

$$q = \frac{1}{a^3}\int_0^a r^3 \sin r\cos r\, dr = \frac{3}{2\pi} - \frac{12}{\pi^3}. \tag{13}$$

Both A and B are strictly positive on the entire noise interval. At zero noise the individual states are pure, while their prior average is full rank.

III. A FINITE ATTAINING MEASUREMENT

Let the four sign vectors be

$$\begin{aligned} \boldsymbol{s}_0 &= (1,1,1), \qquad \boldsymbol{s}_1 = (1,-1,-1), \\ \boldsymbol{s}_2 &= (-1,1,-1), \quad \boldsymbol{s}_3 = (-1,-1,1). \end{aligned} \tag{14}$$

Define the outcome vectors and effects

$$|e_j\rangle = \frac{|0\rangle + \sum_i s_{ji}|\widetilde{i}\rangle}{2}, \qquad M_j = |e_j\rangle\langle e_j|. \tag{15}$$

Since $1 + \boldsymbol{s}_j \cdot \boldsymbol{s}_l = 4\delta_{jl}$, this is an orthonormal projective measurement on the full two-qubit output space. The outcome probabilities and first moments are

$$p_j = \mathrm{Tr}(\Gamma_0 M_j) = \frac{1}{4}, \qquad g_{ij} = \mathrm{Tr}(\Gamma_i M_j) = \frac{Q}{2}s_{ji}. \tag{16}$$

For quadratic loss the optimal estimator at fixed measurement is the posterior mean, so

$$\hat{\boldsymbol{\theta}}_j = \frac{\boldsymbol{g}_j}{p_j} = 2Q\boldsymbol{s}_j. \tag{17}$$

The posterior decomposition of squared error yields

$$R(M, \hat{\boldsymbol{\theta}}) = R_{\mathrm{p}} - \frac{1}{3}\sum_j p_j\|\hat{\boldsymbol{\theta}}_j\|^2 = R_{\mathrm{p}} - 4Q^2. \tag{18}$$

The measurement does not depend on λ; only the estimates do. They remain in the prior ball: from $q = \mathbb{E}[r\sin r\cos r]/3 \le \mathbb{E}[r]/6 = a/8$,

$$\|\hat{\boldsymbol{\theta}}_j\| = 2\sqrt{3}Q \le \frac{\sqrt{3}a}{4} < a. \tag{19}$$

The measurement also has a simple two-qubit implementation. Set

$$V_j = \frac{\mathbb{I}_2 - i\boldsymbol{s}_j\cdot\boldsymbol{\sigma}}{2}, \qquad |e_j\rangle = (V_j \otimes \mathbb{I}_2)|\Phi^+\rangle. \tag{20}$$

Each V_j is unitary. For the stated ordering,

$$V_0^\dagger V_j = \mathbb{I}_2,\ i\sigma_y,\ i\sigma_z,\ i\sigma_x \quad (j = 0,1,2,3). \tag{21}$$

Apply $V_0^\dagger$ to the system after the noisy channel and then perform a Bell-basis measurement, with the label permutation in Eq. (21). This implements Eq. (15). It is a circuit identity requiring a joint Bell measurement, rather than a claim of an experimental implementation or a product measurement.

IV. EXACT OPTIMALITY AND NH SATURATION

OpenAI Codex assisted with the derivations and verification code used here and in the appendices. Human scientific verification and final tool-version disclosure remain pending for this local draft.

For the fixed probe, the Bayesian NH program of Refs. [3, 5] is the minimization of

$$\mathcal{R}(L, X) = R_{\mathrm{p}} + \frac{1}{3}\sum_i \mathrm{Tr}(\Gamma_0 L_{ii}) - \frac{2}{3}\sum_i \mathrm{Tr}(\Gamma_i X_i) \tag{22}$$

over Hermitian X_i and blocks $L_{ij} = L_{ji} = L_{ij}^\dagger$, subject to

$$S(L, X) = \begin{pmatrix} L & X \\ X^\dagger & \mathbb{I}_4 \end{pmatrix} \succeq 0, \qquad X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}. \tag{23}$$

Every physical measurement with real estimates supplies feasible moments $X_i = \int t_i\, dM$ and $L_{ij} = \int t_i t_j\, dM$:

$$S(L, X) = \int \begin{pmatrix} t_1\mathbb{I}_4 \\ t_2\mathbb{I}_4 \\ t_3\mathbb{I}_4 \\ \mathbb{I}_4 \end{pmatrix} dM \begin{pmatrix} t_1\mathbb{I}_4 & t_2\mathbb{I}_4 & t_3\mathbb{I}_4 & \mathbb{I}_4 \end{pmatrix}. \tag{24}$$

Thus its optimum C_{NH} is a lower bound on the physical optimum R_1. To prove tightness, we give a dual matrix independently of a numerical solution.

In $\mathbb{C}^3 \otimes \mathbb{C}^4$, the first factor labels parameters and the second uses Eq. (7). Define

$$
\begin{aligned}
|\omega\rangle &= \sum_{i=1}^{3} |i\rangle \otimes |\widetilde{i}\rangle, \\
|h_{ij}\rangle &= |i\rangle \otimes |\widetilde{j}\rangle - |j\rangle \otimes |\widetilde{i}\rangle, \\
K &= \frac{A}{3} \sum_i |i,0\rangle\langle i,0| + \frac{B}{3} |\omega\rangle\langle\omega| + \frac{B}{3} \sum_{i<j} |h_{ij}\rangle\langle h_{ij}|.
\end{aligned}
\tag{25}
$$

This is positive semidefinite. Its 4×4 Hilbert-space blocks are

$$
K_{ij} = \frac{A}{3}\delta_{ij} P_0 + \frac{B}{3}\left(\delta_{ij} P_1 + E_{ij} - E_{ji}\right). \tag{26}
$$

Let $x = 2Q$ and let $X^\star$ be the column of the three operators $X_i^\star = xF_i$. These operators define the certificate; they need not equal the finite measurement's first moments. The lifted matrix

$$
Z = \begin{pmatrix} \mathbb{I}_{12} \\ -X^{\star\dagger} \end{pmatrix} K \begin{pmatrix} \mathbb{I}_{12} & -X^\star \end{pmatrix} \succeq 0, \tag{27}
$$

$$
\begin{aligned}
(KX^\star)_i &= x\left(\frac{A}{3} E_{0i} + B E_{i0}\right), \\
Z_{44} &= x^2\left(3BP_0 + \frac{A}{3} P_1\right)
\end{aligned}
\tag{28}
$$

obeys, by $A + 3B = 1$,

$$
\begin{aligned}
Z_{ii} &= \Gamma_0/3, \\
Z_{ij} + Z_{ji} &= 0 \quad (i < j \le 3), \\
Z_{i4} + Z_{4i} &= -2\Gamma_i/3, \\
\operatorname{Tr} Z_{44} &= 4Q^2.
\end{aligned}
\tag{29}
$$

For any feasible $S(L, X)$, these equations give the exact weak-duality identity

$$
\mathcal{R}(L, X) - (R_{\mathrm{p}} - 4Q^2) = \operatorname{Tr}[Z S(L, X)] \ge 0. \tag{30}
$$

The off-diagonal K contributions cancel because $L_{ij} = L_{ji}$; the Hermitian X_i give precisely the first-moment coefficients in Eq. (22). Consequently

$$
R_{\mathrm{p}} - 4Q^2 \le C_{\mathrm{NH}} \le R_1 \le R(M, \hat{\boldsymbol{\theta}}) = R_{\mathrm{p}} - 4Q^2. \tag{31}
$$

All inequalities are equalities, proving the stated result:

$$
\boxed{
\begin{aligned}
R_1(\lambda) &= C_{\mathrm{NH}}(1, \lambda) \\
&= \frac{\pi^2}{80} - 4(1-\lambda)^2 \left(\frac{3}{2\pi} - \frac{12}{\pi^3}\right)^2 .
\end{aligned}
}
\tag{32}
$$

Positivity follows from the Gram factors in Eqs. (25) and (27), so no solver tolerance or noise grid enters the proof. A separate physical loss-operator certificate, valid for every real estimate, is given in Appendix A.

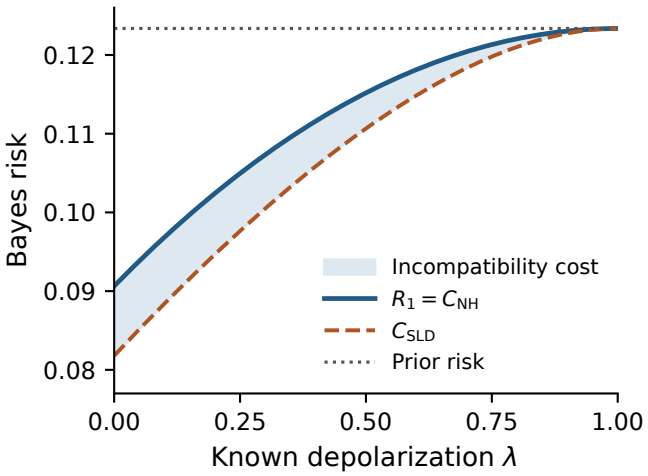


FIG. 1. Dimensionless mean-square coordinate risk for one fixed Bell probe and the radius-$\pi/4$ volume prior. The optimal risk and NH bound are the same solid curve, Eq. (32); the dashed curve is the Bayesian SLD bound, Eq. (35); the dotted line is the prior risk. The shaded separation is the exact incompatibility penalty. The curves are evaluations of analytic formulas, not measurement data or numerical optimization results. OpenAI Codex assisted with the plotting code; the figure was rendered from the accompanying formula table.

V. COVARIANCE AND THE INCOMPATIBILITY PENALTY

An alternative optimal measurement is continuous. With the normalized sphere measure $d\mu(\boldsymbol{n}) = d\Omega/(4\pi)$, set

$$
\begin{aligned}
|m(\boldsymbol{n})\rangle &= |0\rangle + \sqrt{3} \sum_i n_i |\widetilde{i}\rangle, \\
dM(\boldsymbol{n}) &= |m(\boldsymbol{n})\rangle\langle m(\boldsymbol{n})|\, d\mu(\boldsymbol{n}).
\end{aligned}
\tag{33}
$$

Spherical first and second moments imply $\int dM = \mathbb{I}_4$. The outcome probability density relative to $d\mu$ is $A + 3B = 1$, and the posterior mean is $2\sqrt{3}Q\boldsymbol{n}$. Its risk is therefore again $R_{\mathrm{p}} - 4Q^2$. This POVM is covariant under the scalar and vector representation of SO(3). The four-outcome measurement is a finite optimal alternative; it does not discretize the continuous parameter prior.

The finite measurement's moment operators make its mechanism explicit. For $\{i, j, l\} = \{1, 2, 3\}$,

$$
X_i^{\mathrm{tet}} = 2Q(F_i + E_{jl} + E_{lj}), \qquad (X_i^{\mathrm{tet}})^2 = 4Q^2 \mathbb{I}_4. \tag{34}
$$

They commute pairwise. The triplet completion terms have zero overlap with the signal first moments Γ_i and enable a common projective measurement. They do not turn the SLD operators into commuting operators. Indeed the Bayesian Lyapunov equations $(\Gamma_0 S_i + S_i \Gamma_0)/2 =$

Γ_i give

$$S_i = \frac{2Q}{A+B}F_i,$$
$$C_{\mathrm{SLD}} = R_{\mathrm{p}} - \frac{4Q^2}{A+B},$$
$$R_1 - C_{\mathrm{SLD}} = \frac{8BQ^2}{A+B} > 0 \quad (\lambda < 1). \tag{35}$$

This is a finite Bayesian incompatibility cost, rather than an asymptotic or locally unbiased variance. Figure 1 displays these exact expressions.

At zero noise, the formulas give

$$R_1(0) \simeq 0.09064783829,$$
$$C_{\mathrm{SLD}}(0) \simeq 0.08180546613,$$
$$R_1(0)/C_{\mathrm{SLD}}(0) \simeq 1.108090236. \tag{36}$$

This reproduces the rounded single-use ratio 1.108 reported in Ref. [5]. For one use, rotation averaging of a tester preserves the isotropic prior and loss, and irreducibility of the qubit input representation fixes its invariant marginal to $\mathbb{I}_2/2$; hence the source's invariant single-use benchmark maps to this Bell experiment. This benchmark comparison does not assert optimality of Bell probes for multiple uses.

VI. SCOPE AND PHYSICAL INTERPRETATION

The optimal apparatus can be fixed while the known depolarization changes: the posterior vectors are multiplied by $1-\lambda$, and the precision gain below the prior risk scales as $(1-\lambda)^2$. At $\lambda = 0$ the proof applies directly to pure individual outputs, using positive prior moments and exact Gram factors. At $\lambda = 1$, $Q = 0$ and all estimates are zero; the physical optimum, NH bound, and SLD bound are the prior risk. No full-rank endpoint continuation, perturbative expansion, or large-sample approximation is needed.

The closest phase-imaging moments in Ref. [4], Eqs. (58)–(59), have the same star structure after rephasing: for three parameters, $\alpha^2 = A/B$ and a first-moment rescaling match the measurement optimization, while the prior-loss constants differ. Thus the star reduction is not an additional contribution of this work. The finite measurement and positive NH certificate supply the exact attainment step at the present rotation-model scope.

Although the source numerical study extends through four parallel uses, this paper concerns one use with a fixed Bell probe. In a multiple-use experiment every output encodes the same unknown vector, so $\int p(\boldsymbol{\theta})\rho_{\lambda,\boldsymbol{\theta}}^{\otimes k} d^3\boldsymbol{\theta}$ is generally not $\Gamma_0^{\otimes k}$. The present four-dimensional certificates therefore do not settle NH attainment for $k = 2, 3, 4$. They also do not establish a global optimum over arbitrary probes or adaptive protocols.

DATA AVAILABILITY

The analytic expressions and optimality certificates are given in this article. Symbolic checks, direct measurement-replay code, and the code and tabulated formula evaluations for Fig. 1 accompany the local draft. No experimental data were used.

ACKNOWLEDGMENTS AND USE OF ARTIFICIAL INTELLIGENCE

OpenAI Codex assisted with literature checks, analytic derivations, symbolic and numerical verification code, figure generation, and manuscript preparation. Automated checks included a separately reconstructed calculation in a fresh AI context. Human scientific review and approval of this local draft, including confirmation of tool versions and the final disclosure, remain pending.

Appendix A: A physical loss-operator certificate

Define the second-moment loss operator

$$H = \frac{1}{3}\int p(\boldsymbol{\theta})\|\boldsymbol{\theta}\|^2 \rho_{\lambda,\boldsymbol{\theta}}\, d^3\boldsymbol{\theta}, \qquad \mathrm{Tr}\, H = R_{\mathrm{p}}. \tag{A1}$$

For a reported vector $\boldsymbol{t} \in \mathbb{R}^3$, its loss operator is

$$\mathcal{F}(\boldsymbol{t}) = H + \frac{1}{3}\left(\|\boldsymbol{t}\|^2\Gamma_0 - 2\sum_i t_i\Gamma_i\right). \tag{A2}$$

The general supporting-operator criterion is part of Bayesian quantum decision theory; see Appendix B of Ref. [4]. Here a specific supporting operator is

$$Y = H - 12BQ^2P_0 - \frac{4}{3}AQ^2P_1. \tag{A3}$$

Rotational covariance lets us take $\boldsymbol{t} = u\boldsymbol{e}_3$, where $u = \|\boldsymbol{t}\| \geq 0$. The $0,3$ block of $\mathcal{F}(\boldsymbol{t}) - Y$ is

$$\frac{1}{3}\begin{pmatrix} Au^2 + 36BQ^2 & -2Qu \\ -2Qu & Bu^2 + 4AQ^2 \end{pmatrix}; \tag{A4}$$

the two transverse entries are $(Bu^2 + 4AQ^2)/3$. All diagonal entries are nonnegative, and the block determinant is

$$\det[\mathcal{F}(\boldsymbol{t}) - Y]_{0,3} = \frac{AB}{9}(u^2 - 12Q^2)^2 \geq 0. \tag{A5}$$

Thus $\mathcal{F}(\boldsymbol{t}) \succeq Y$ for every real estimate, including estimates outside the prior ball. Integrating against any POVM proves $R \geq \mathrm{Tr}\, Y = R_{\mathrm{p}} - 4Q^2$. At $u = 2\sqrt{3}Q$ the block annihilates $(1, \sqrt{3})^T$, giving equality for the sphere measurement and the four tetrahedral directions. This certificate proves physical optimality without assuming that every feasible NH moment pair is realizable.

Appendix B: Explicit covariant NH primal

For completeness, the sphere measurement supplies

$$X_i^{\mathrm{sph}} = 2QF_i,$$
$$L_{ij}^{\mathrm{sph}} = 4Q^2\delta_{ij}P_0 + \frac{12Q^2}{5}(\delta_{ij}P_1 + E_{ij} + E_{ji}). \quad \text{(B1)}$$

These formulas follow from $\mathbb{E}[n_i n_j n_a n_b] = (\delta_{ij}\delta_{ab} + \delta_{ia}\delta_{jb} + \delta_{ib}\delta_{ja})/15$. The Schur complement $L^{\mathrm{sph}} - X^{\mathrm{sph}}X^{\mathrm{sph}\dagger}$ vanishes on the scalar Hilbert sector. On the parameter–triplet tensor space it vanishes on trace and antisymmetric tensors and equals $24Q^2/5$ times the identity on symmetric traceless tensors. This gives another direct positive-semidefinite primal with objective $R_{\mathrm{p}} - 4Q^2$, including $Q = 0$.

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